

MAM2000W
Linear Transformations

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Linear Transformations

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Chapter 1

Linear Transformations

1.1 Linear Transformations

Definition 1.1.1

Let T be a function from vector space V to vector space W . We say that T is a linear transformation if it satisfies 2 conditions:

- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for every $\mathbf{u}, \mathbf{v} \in V$
- $T(\lambda \mathbf{u}) = \lambda T(\mathbf{u})$ for every scalar λ and vector $\mathbf{u} \in V$

Example 1.1.1

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & -3 & 2 \end{pmatrix}$. For every $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3$, let

$$T(\mathbf{x}) = A\mathbf{x} = \begin{pmatrix} x_1 + 2x_2 + 3x_3 \\ x_1 - 3x_2 + 2x_3 \end{pmatrix}$$

This T defines a function from vector space $\mathbb{R}^3$ to $\mathbb{R}^2$. Since we know that $A(B + C) = AB + AC$ and $A(\lambda B) = \lambda(AB)$ for matrices A, B, C , we see that T is a linear transformation

There was nothing important about the matrix A , we chose. In fact, the following is true

Theorem 1.1.1

If A is an $m \times n$ matrix, then the function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(\mathbf{u}) = A\mathbf{u}$ for every $\mathbf{u} \in \mathbb{R}^n$ is a linear transformation

Note:-

You probably already came across the idea that one could think of a matrix as a special type of function in first year mathematics. We now know what is special about them: they are linear functions. In particular, you saw there that the geometrical operations of reflection, projection and rotation in the plane (i.e. in $\mathbb{R}^2$) can be represented by using matrices. We now know that they are in fact linear transformations.

Example 1.1.2

The matrices

$$\text{Rot}_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \text{Proj}_x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \text{Ref}_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

represent respectively a rotation through an angle of θ counter-clockwise about the origin, a projection

onto the x -axis and a reflection in the x -axis

Note:-

We have that $f(x) = mx$ is linear, but $f(x) = mx + c$ is not. We should actually call $f(x) = mx + c$ affine functions, not linear functions

Note:-

To show $T : V \rightarrow W$ is not linear, it is sufficient to find either:

- 2 vectors $\mathbf{u}, \mathbf{v} \in V$ such that $T(\mathbf{u} + \mathbf{v}) \neq T(\mathbf{u}) + T(\mathbf{v})$
- A vector $\mathbf{u} \in V$ and scalar λ such that $T(\lambda\mathbf{u}) \neq \lambda T(\mathbf{u})$

Theorem 1.1.2

Let $T : V \rightarrow W$ be a linear transformation, $\mathbf{u}, \mathbf{v}, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n \in V$ and $\lambda_1, \lambda_2, \dots, \lambda_n$ scalars. Then

1. $T(\lambda_1\mathbf{u}_1 + \lambda_2\mathbf{u}_2 + \dots + \lambda_n\mathbf{u}_n) = \lambda_1T(\mathbf{u}_1) + \lambda_2T(\mathbf{u}_2) + \dots + \lambda_nT(\mathbf{u}_n)$
2. $T(\mathbf{0}) = \mathbf{0}$
3. $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$

Proof:

1. This uses a fairly simple induction
2. Since $\mathbf{0} + \mathbf{0} = \mathbf{0}$, we get that $T(\mathbf{0} + \mathbf{0}) = T(\mathbf{0}) \implies T(\mathbf{0}) + T(\mathbf{0}) = T(\mathbf{0})$, then adding the negative of $T(\mathbf{0})$ to both sides yields $T(\mathbf{0}) = \mathbf{0}$

Note:-

The two zero vectors on either side may be different. The left zero vector is the zero vector in V whereas the right is in W . So we actually have that $T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) \implies T(\mathbf{0}_V) + T(\mathbf{0}_V) = T(\mathbf{0}_V)$. Now $T(\mathbf{0}_V) \in W \implies T(\mathbf{0}_V) = \mathbf{0}_W$

3. Remember that $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}) = \mathbf{u} + (-1)\mathbf{v}$ and then split as normal

☺

Note:-

We can use the second property to show a function is not linear

Corollary 1.1.1

Let V and W be vector spaces and $F : V \rightarrow W$ a function. If $F(\mathbf{0}_V) \neq \mathbf{0}_W$, then F is not a linear transformation

Example 1.1.3

$F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $F\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + 1 \\ x_2 - 2 \end{pmatrix}$ is not a linear transformation because $F\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \neq \mathbf{0}$

Note:-

In the above case, we assume the vector spaces is the usual addition and multiplication for that set. Also if $F(\mathbf{0}) = \mathbf{0}$, then F is not necessarily a linear transformation.

Note:-

Suppose V is an n -dimensional vector space, $T : V \rightarrow W$ is a linear transformation and $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V . Then let $\mathbf{v} \in V$ be any vector. Since B is a basis for V , we know that there are scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ such that

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$$

Then, by the last theorem, $T(\mathbf{v}) = T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n) = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) + \dots + \lambda_n T(\mathbf{v}_n)$. Thus if we know $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ we can find $T(\mathbf{v})$ for any $\mathbf{v} \in V$. This shows that once we know what a linear transformation "does" to the vectors in a basis for its domain, we know what it "does" to every vector in its domain

Example 1.1.4

Suppose T is linear from P_1 to P_2 and that $T(x+1) = x^2 - 1$ and $T(x-1) = x^2 + x$. We wish to work out $T(ax+b)$. First we see that $\{x-1, x+1\}$ is a basis for P_1 . Then $ax+b = \frac{a-b}{2}(x-1) + \frac{a+b}{2}(x+1)$ so $T(ax+b) = \frac{a-b}{2}T(x-1) + \frac{a+b}{2}T(x+1) = \frac{a-b}{2}(x^2+x) + \frac{a+b}{2}(x^2-1)$

1.2 The Kernel and Range of a Linear Transformation

Definition 1.2.1

Let $T : V \rightarrow W$ be a linear transformation. The kernel of T is the subset of V consisting of all vectors $\mathbf{u} \in V$ such that $T(\mathbf{u}) = \mathbf{0}$. It is denoted by $\ker(T) = \{\mathbf{u} \in V : T(\mathbf{u}) = \mathbf{0}\}$

Example 1.2.1

Let A be an $m \times n$ matrix and T be the associated linear transformation from $\mathbb{R}^n \rightarrow \mathbb{R}^m$ [$T(\mathbf{u}) = A\mathbf{u}$]
 $\ker(T) = \{\mathbf{u} \in \mathbb{R}^n : A\mathbf{u} = \mathbf{0}\} = NS(A)$

In this case, we see that the kernel of a linear transformation is a subspace. In fact, this is true in general

Theorem 1.2.1

If V and W are vector spaces and $T : V \rightarrow W$ is a linear transformation, then $\ker(T)$ is a subspace of V

Proof: We have that $T(\mathbf{0}) = \mathbf{0}$ so $\mathbf{0} \in \ker(T)$. Now suppose $\mathbf{u}, \mathbf{v} \in \ker(T)$. We wish to show that $\mathbf{u} + \mathbf{v} \in \ker(T)$. Now we have that $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{0} + \mathbf{0} = \mathbf{0} \implies \mathbf{u} + \mathbf{v} \in \ker(T)$. Also $T(\lambda\mathbf{u}) = \lambda T(\mathbf{u}) = \lambda\mathbf{0} = \mathbf{0}$ so $\ker(T)$ is a subspace of V $\oplus$

Example 1.2.2

For $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined as $(x, y, z) \mapsto (x+y, x-z)$, we get that

$$\begin{aligned} \ker(T) &= \{(x, y, z) : T(x, y, z) = (0, 0)\} \\ &= \{(x, y, z) : (x+y, x-z) = (0, 0)\} \\ &= \{(x, y, z) : x = t, y = -t, z = t, \text{ for } t \in \mathbb{R}\} = \{\lambda(1, -1, 1) : \lambda \in \mathbb{R}\} \end{aligned}$$

So we see that a basis for $\ker(T) = \{(1, -1, 1)\}$ and $\dim \ker(T) = 1$

Theorem 1.2.2

If V and W are vector spaces and $T : V \rightarrow W$ is a linear transformation, then T is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$

Proof: Suppose T is one-to-one and $\mathbf{v} \in \ker(T)$. We wish to show that $\mathbf{v} = \mathbf{0}$. Since $\mathbf{v} \in \ker(T)$, we get that $T(\mathbf{v}) = \mathbf{0}$. But we know that $T(\mathbf{0}) = \mathbf{0}$, so because T is injective, we get that $\mathbf{v} = \mathbf{0}$. Conversely, assume $\ker(T) = \{\mathbf{0}\}$ and that $T(\mathbf{u}) = T(\mathbf{v})$. We wish to show that $\mathbf{u} = \mathbf{v}$. Since $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$, we get that $\mathbf{u} - \mathbf{v} \in \ker(T) \implies \mathbf{u} - \mathbf{v} = \mathbf{0} \implies \mathbf{u} = \mathbf{v}$ $\odot$

Definition 1.2.2

Let $T : V \rightarrow W$ be a linear transformation, then the set $\{T(\mathbf{v}) : \mathbf{v} \in V\}$ is called the range of T and is denoted by $\text{ran}(T)$ (sometimes written as $T^\rightarrow(V)$ or $T(V)$). The transformation is surjective if $\text{ran}(T) = W$

Theorem 1.2.3

Let $T : V \rightarrow W$ be a linear transformation, then $\text{ran}(T)$ is a subspace of W

Proof: We have that $\mathbf{0} \in \text{ran}(T)$, because $T(\mathbf{0}) = \mathbf{0}$. Now if $\mathbf{w}_1, \mathbf{w}_2 \in \text{ran}(T)$, then we have that $\mathbf{w}_1 = T(\mathbf{v}_1)$ and $\mathbf{w}_2 = T(\mathbf{v}_2)$ for some $\mathbf{v}_1, \mathbf{v}_2 \in V$. Thus $\mathbf{w}_1 + \mathbf{w}_2 = T(\mathbf{v}_1) + T(\mathbf{v}_2) = T(\mathbf{v}_1 + \mathbf{v}_2)$ so $\mathbf{w}_1 + \mathbf{w}_2 \in \text{ran}(T)$. Similarly, we have that $\lambda \mathbf{w}_1 = \lambda T(\mathbf{v}_1) = T(\lambda \mathbf{v}_1)$ so $\lambda \mathbf{w}_1 \in \text{ran}(T)$ $\odot$

Example 1.2.3

Consider $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T(\mathbf{v}) = A\mathbf{v}$ where A is some $m \times n$ matrix. Then $\text{ran}(T) = \{\mathbf{w} \in \mathbb{R}^m : \mathbf{w} = A\mathbf{v} \text{ for some } \mathbf{v} \in \mathbb{R}^n\}$ which gives $\text{ran}(T) = CS(A)$

Claim 1.2.1

$\text{ran}(T) = CS(A)$ in the above example

Proof: Let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ be the columns of A and $\mathbf{v} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}$. Now $A\mathbf{v} \in \text{ran}(T)$ if and only if $A\mathbf{v} = (\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n) \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} = \lambda_1 \mathbf{c}_1 + \lambda_2 \mathbf{c}_2 + \dots + \lambda_n \mathbf{c}_n$ if and only if $A\mathbf{v}$ is a linear combination of the columns of A if and only if $A\mathbf{v} \in CS(A)$.

Note:-

This is still an $m \times 1$ matrix because each column vector is an $m \times 1$ matrix $\odot$

Theorem 1.2.4

Let $T : V \rightarrow W$ be a linear transformation and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ a basis for V , then $\text{ran}(T) = \text{span}(\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\})$

Proof: Each $T(\mathbf{v}_i) \in \text{ran}(T)$, so $\text{span}(\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}) \subseteq \text{ran}(T)$. Now $\mathbf{w} \in \text{ran}(T)$ so $\mathbf{w} = T(\mathbf{v})$ and since $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ form a basis for V , we get that $\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$ for some $\lambda_1, \lambda_2, \dots, \lambda_n \in F$. Then we get that $\mathbf{w} = T(\mathbf{v}) = T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n) = T(\lambda_1 \mathbf{v}_1) + T(\lambda_2 \mathbf{v}_2) + \dots + T(\lambda_n \mathbf{v}_n)$ so $\mathbf{w} \in \text{span}(\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}) \implies \text{ran}(T) \subseteq \text{span}(\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}) \implies \text{ran}(T) = \text{span}(\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\})$ $\odot$

Example 1.2.4

Find basis for $\text{ran}(T)$ with $T : P_3 \rightarrow P_3$ given by $T(a_0 + a_1x + a_2x^2 + a_3x^3) = (a_1 - a_0)x + (a_3 - a_2)x^3$

Consider basis $1, x, x^2, x^3$ for P_3

Then use $\text{ran}(T) = \text{span}(\{T(1), T(x), T(x^2), T(x^3)\})$ but $T(1)$ and $T(x)$ are linearly dependent, and $T(x^2)$ and $T(x^3)$ are linearly dependent, so we can take x, x^3 as basis for $\text{ran}(T)$

(see notes)

Proposition 1.2.1

Let $T : V \rightarrow W$ be a bijective linear transformation, then inverse function $T^{-1} : W \rightarrow V$ is also a linear transformation

Proof: $T^{-1}(\mathbf{w} + \mathbf{v}) = T^{-1}(T(T^{-1}(\mathbf{w}) + T^{-1}(\mathbf{v}))) = T^{-1}(T(T^{-1}(\mathbf{w}))) + T^{-1}(T(T^{-1}(\mathbf{v}))) = T^{-1}(\mathbf{w}) + T^{-1}(\mathbf{v})$

We wish to show that $T^{-1}(\lambda \mathbf{w}) = \lambda T^{-1}(\mathbf{w}), \forall \mathbf{w} \in W$. So suppose $T^{-1}(\mathbf{w}) = \mathbf{v} \iff T(\mathbf{v}) = \mathbf{w}$ for $\mathbf{v} \in V$. Then $T^{-1}(\lambda \mathbf{w}) = T^{-1}(\lambda T(\mathbf{v})) = T^{-1}(T(\lambda \mathbf{v})) = \lambda \mathbf{v} = \lambda T^{-1}(\mathbf{w})$ ☺

Definition 1.2.3

Let V, W be vector spaces (over F), a bijective linear transformation $T : V \rightarrow W$ is called a linear isomorphism, and we call V and W isomorphic [if there exists some isomorphism, then V and W are isomorphic]

Example 1.2.5

- P_n (polynomials of degree at most n) and $\mathbb{R}^{n+1}$ are isomorphic because $T : P_n \rightarrow \mathbb{R}^{n+1}$ defined by $T(a_0 + a_1x + \dots + a_nx^n) = (a_0, a_1, \dots, a_n)$ is a linear isomorphism
- $M_{2 \times 2}(\mathbb{R})$ and $\mathbb{R}^4$ are isomorphic. Define isomorphism $T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^4$ by $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (a, b, c, d)$

Note:-

We check this is a linear transformation and that it is bijective

- Consider $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ with P_x defined as the projection to the x -axis, meaning $P_x(x, y) = (x, 0)$ and R_x is the reflection over x -axis, so $R_x(x, y) = (x, -y)$. We have that P_x is linear but not bijective and R_x is a linear isomorphism (which is self inverse)

Note:-

This is called a linear transformation because it is a function from one vector space to another that respects the underlying (linear) structure of each vector space

1.3 The Matrix of a Linear Transformation

Definition 1.3.1

Let V be an n -dimensional vector space with basis B and $\mathbf{v} \in V$. We write $[\mathbf{v}]_B$ for the coordinate vector of $\mathbf{v}$ with respect to the (ordered) basis B

Example 1.3.1

For $B_1 = \{(1, 0), (0, 1)\}$ and $B_2 = \{(1, 1), (1, -1)\}$, we have that $[(3, 5)]_{B_1} = (3, 5)$ and $[(3, 5)]_{B_2} = (4, -1)$

Theorem 1.3.1

Let B be a basis for vector space V and $\mathbf{v}_1, \mathbf{v}_2 \in V$ and λ a scalar. Then

- $[\mathbf{v}_1 + \mathbf{v}_2]_B = [\mathbf{v}_1]_B + [\mathbf{v}_2]_B$
- $[\lambda \mathbf{v}_1]_B = \lambda [\mathbf{v}_1]_B$

Proof: You will need to use the fact that every vector in a vector space has a unique representation in terms of the basis vectors. If you can write down two representations for the vector $\mathbf{v}_1 + \mathbf{v}_2$ for example, they must be equal. ☺

Corollary 1.3.1

Let B be a basis for an n -dimensional real vector space V . Then the function $F : V \rightarrow \mathbb{R}^n$ defined by $F(\mathbf{v}) = [\mathbf{v}]_B$ is a linear isomorphism. This means every n -dimensional real vector space is isomorphic to the vector space $\mathbb{R}^n$. Similarly, every complex vector space is isomorphic to $\mathbb{C}^n$

Proof: Theorem above shows that F is linear. It is an easy exercise, using the fact that B is a basis, to show that F is bijective $\odot$

We now prove the main result:

Theorem 1.3.2

Let $T : V \rightarrow W$ be a linear transformation and suppose V is an n -dimensional space and W is m -dimensional. Thus let $B_V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for V and $B_W = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ a basis for W . Then create a matrix A whose j th column is the coordinate vector $[T(\mathbf{v}_j)]_{B_W}$. Then A is an $m \times n$ matrix with the following property:

$$A[\mathbf{v}]_{B_V} = [T(\mathbf{v})]_{B_W} \text{ for any vector } \mathbf{v} \in V$$

Moreover A is unique (the only matrix with this property)

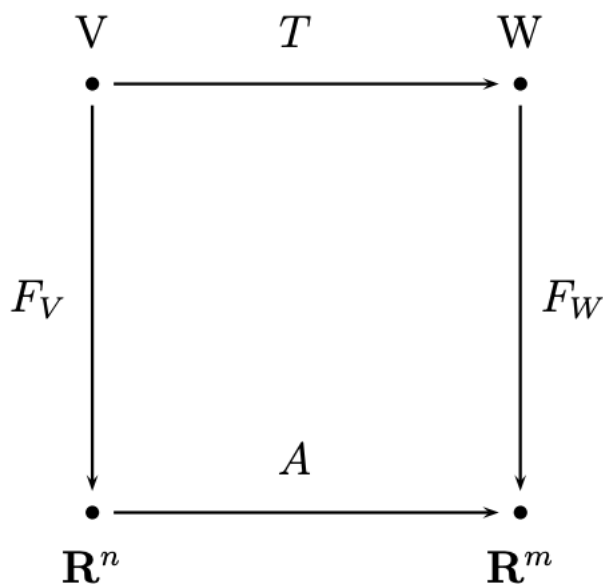
Note:-

We say that A is the matrix which represents T with respect to the two bases B_V and B_W . We denote ${}_{B_W}[T]_{B_V}$. If $V = W$ and we use the same basis B for V both times, A is called the matrix which represents T with respect to B

If functions $F_V : V \rightarrow \mathbb{R}^n$ and $F_W : W \rightarrow \mathbb{R}^m$ are defined by $F_V(\mathbf{v}) = [\mathbf{v}]_{B_V}$ and $F_W(\mathbf{w}) = [\mathbf{w}]_{B_W}$, then it follows that they are both isomorphisms and the equation from the above formula becomes

$$A(F_V(\mathbf{v})) = F_W(T(\mathbf{v}))$$

The diagram below gives one way of picturing this equation.



Here are a few examples before proving it

Example 1.3.2

Let $T : P_1 \rightarrow P_2$ be the linear transformation defined by $T(p(x)) = x(p(x))$

Let us find the matrix which represents T with respect to bases $B_1 = \{1, x\}$ and $B_2 = \{1, x, x^2\}$
 We need to calculate $[T(1)]_{B_2}$ and $[T(x)]_{B_2}$

$$[T(1)]_{B_2} = [x]_{B_2} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } [T(x)]_{B_2} = [x^2]_{B_2} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{Then } A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

To check, we will calculate $[T(ax + b)]_{B_2}$ and $A[ax + b]_{B_1}$ separately (we should get the same answer)

$$\text{Firstly } T(ax + b) = x(ax + b) = ax^2 + bx \implies [T(ax + b)]_{B_2} = \begin{pmatrix} 0 \\ b \\ a \end{pmatrix}$$

$$\text{Secondly, } A[ax + b]_{B_1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ b \\ a \end{pmatrix}$$

Now we prove this theorem:

Proof: Suppose $[\mathbf{v}]_{B_V} = (\lambda_1, \lambda_2, \dots, \lambda_n)$, then

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$$

By the linearity of T , we get that:

$$T(\mathbf{v}) = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) + \dots + \lambda_n T(\mathbf{v}_n)$$

$$\begin{aligned} [T(\mathbf{v})]_{B_W} &= \lambda_1 [T(\mathbf{v}_1)]_{B_W} + \lambda_2 [T(\mathbf{v}_2)]_{B_W} + \dots + \lambda_n [T(\mathbf{v}_n)]_{B_W} \\ &= A \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} \end{aligned}$$

since A has $[T(\mathbf{v}_j)]_{B_W}$ for its columns and thus $[T(\mathbf{v})]_{B_W} = A[\mathbf{v}]_{B_V}$

Now we have that $[T(\mathbf{v})]_{B_W} =$

⊖

Example 1.3.3

Example 3.3.7 Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. Then, we know from the previous section that we can think of A as a linear mapping from $\mathbf{R}^3$ to $\mathbf{R}^2$. It is a natural to ask at this stage what the matrix representing A with respect to the usual bases for $\mathbf{R}^3$ and $\mathbf{R}^2$ is.

The theorem says we should apply the mapping to the basis vectors chosen in $\mathbf{R}^3$. We get

$$A \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

(Since we are working with the usual bases in $\mathbf{R}^3$ and $\mathbf{R}^2$ the coordinate vector of any vector $\mathbf{v}$ is just $\mathbf{v}$ itself.) Thus the matrix representing A with respect to the usual bases is just A itself. We could make this a general theorem, but it's pretty obvious from this example.

Example 3.3.8 Let A be the matrix of the previous example. We find the matrix ${}_{B_2}T_{B_1}$ which represents A with respect to the bases

$$B_1 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\} \quad \text{and} \quad B_2 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

for $\mathbf{R}^3$ and $\mathbf{R}^2$ respectively.

The theorem says we must calculate the coordinate vectors of

$$A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

with respect to the basis B_2 . We get

$$A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}.$$

We must now find the coordinate vectors of these vectors with respect to B_2 . They are, in order,

$$\begin{pmatrix} \frac{5}{2} \\ -\frac{3}{2} \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}, \quad \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$$

Thus the desired matrix must be

$$\begin{pmatrix} \frac{5}{2} & \frac{1}{2} & 4 \\ -\frac{3}{2} & -\frac{1}{2} & -1 \end{pmatrix}.$$

You should check the answer by trying it out on a few coordinate vectors of some vectors in $\mathbf{R}^3$.

Theorem 1.3.3

Let $T : V \rightarrow W$ be a linear transformation and let V be finite dimensional. Then

$$\dim(\ker(T)) + \dim(\text{ran}(T)) = \dim V$$

Proof: Add in after (in tut) ☺

Corollary 1.3.2

Let V be a finite dimensional vector space and $T : V \rightarrow V$ be a linear transformation. Then the following are equivalent:

- T is injective
- T is surjective
- T is a linear isomorphism

Proof: Exercise. Use the facts that T is injective if and only if $\ker(T) = \{0\}$ and that for a subspace U of V we have $\dim U = \dim V$ if and only if $U = V$. ☺

Note:-

The above result shows that a linear function from a vector space into itself is rather special: if it is either injective or surjective, then it will automatically be both injective and surjective. Convince your self that this need not be true for non-linear functions.